

Exploratory Permutation Testing for Treatment Effect Heterogeneity with XGBoost

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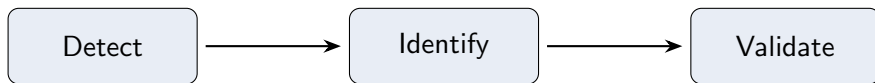
Drug development increasingly asks for whom the treatment works

Patients rarely respond to treatment uniformly. Understanding *who* benefits, and by how much, can:

- reveal **new indications**: a drug that fails on average may work well in a specific subpopulation;
- support **more precise treatment strategies**: matching patients to the therapies most likely to help them; and
- surface **safety signals** in subgroups an average-effect analysis would miss.

Emrelis, an AbbVie antibody-drug conjugate, received accelerated FDA approval in 2025 for non-squamous non-small-cell lung cancer specifically in patients with high c-Met protein expression (Camidge et al. 2024).

Subgroup analysis moves from detecting heterogeneity to validating a clinical claim



- **Detect Heterogeneity:** does the treatment effect vary across patients?
- **Identify Candidate Drivers:** which biomarkers or baseline characteristics contribute to treatment-effect variation, and how?
- **Validate Findings:** does the observed pattern replicate, and support a meaningful treatment-selection strategy?

This talk focuses on testing for treatment-effect heterogeneity.

We study treatment-effect heterogeneity in a two-arm clinical trial

Consider a randomized controlled trial with:

- $A \in \{0, 1\}$: treatment assignment;
- $X \in \mathbb{R}^p$: baseline patient characteristics; and
- Y : clinical outcome measured after treatment (continuous, time-to-event, or binary).

Let $Y(1)$ and $Y(0)$ denote the potential outcomes under treatment and control.

Most of what we will discuss can be applied to observational studies as well, with estimation of the propensity score.

Treatment-effect heterogeneity is variation in the treatment–effect contrast

A **conditional treatment-effect contrast** $\tau(x)$ compares outcomes under treatment and control for a specific subgroup $X = x$.

A common choice is the conditional average treatment effect (CATE):

$$\tau_{\text{CATE}}(x) = E[Y(1) - Y(0) \mid X = x].$$

Other scales define different contrasts, like the conditional risk ratio (binary outcome):

$$\tau_{\text{RR}}(x) = \frac{P[Y(1) = 1 \mid X = x]}{P[Y(0) = 1 \mid X = x]}.$$

Treatment-effect heterogeneity means the chosen contrast $\tau(x)$ varies across x .

Our goal is to test for global treatment-effect heterogeneity

Our goal is to test

$$H_0 : \tau(x) = c \quad \text{for every } x \quad \text{vs.} \quad H_1 : \tau(x) \text{ varies with } x.$$

The null hypothesis is that the treatment effect is *constant* across patient subgroups, not that there is no treatment effect.

Studies are often powered to detect an average treatment effect, so global test results should be viewed as exploratory.

XGBoost can represent complex treatment-effect patterns, but complicates inference

Treatment-effect heterogeneity may involve:

nonlinear
relationships

covariate
interactions

threshold effects

unknown patient
subgroups

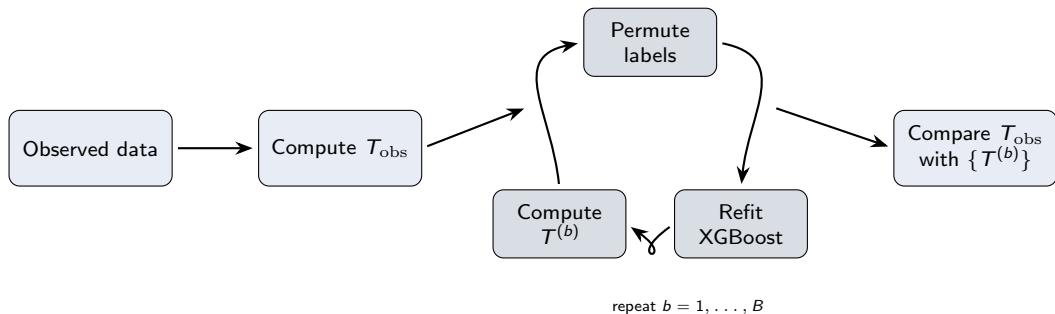
XGBoost (T. Chen and Guestrin 2016) is flexible, representing a fitted function as a sequential regression tree ensemble:

$$f(X) = \sum_{t=1}^T h_t(X).$$

But testing is complicated: the adaptive tree-fitting process makes any resulting test statistic's null distribution hard to characterize.

Permutation testing can calibrate the entire XGBoost procedure

For a statistic T , a permutation procedure constructs the reference distribution by refitting the model:



Because the model is refit for every permutation, the reference distribution reflects the same modeling procedure used to calculate T_{obs} .

Applying a permutation test to ordinary XGBoost leaves two gaps

Permuting treatment labels is useful because it can reproduce the full model-fitting procedure under a no-effect null. However, ordinary XGBoost leaves two gaps:

Target gap

Ordinary XGBoost learns

$$E(Y | X),$$

rather than the treatment-effect contrast $\tau(x)$.

Null gap

Permutation directly represents

$$H_0 : Y_i(1) = Y_i(0),$$

but we care about the broader null

$$H_0 : \tau(x) = c.$$

A-learning helps us target treatment-effect modification

A-learning (S. Chen et al. 2017) models the outcome through a link function g :

$$g\{E(Y | A, X)\} = \underbrace{m(X)}_{\text{baseline}} + \{A - \pi(X)\}f(X),$$

where $\pi(X) = P(A = 1 | X)$ is the propensity score, and $f(X)$ is a "benefiting score".

Since $E\{A - \pi(X) | X\} = 0$, $A - \pi(X)$ is orthogonal to any function of X alone.

So $m(X)$ never enters the estimating problem for $f(X)$, and need not be modeled.

With an identity link, $f(x) = \tau_{\text{CATE}}(x)$; for other links, $f(x)$ may not equal $\tau(x)$.

XGBoostML incorporates a custom A-learning loss in the XGBoost framework

XGBoostML (Liu et al. 2024) represents the benefiting score as a sum of regression trees,

$$f(X) = \sum_{t=1}^T h_t(X),$$

and fits it by minimizing the A-learning-modified loss

$$\mathcal{L}_A(f) = \frac{1}{n} \sum_{i=1}^n M[Y_i, \{A_i - \pi(X_i)\}f(X_i)] + \sum_{t=1}^T \Omega(h_t),$$

where $\Omega(h_t)$ penalizes tree complexity.

Any loss M with $M_v(y, v)$ increasing in v , and $M_v(y, 0)$ monotone in y , gives a valid benefiting score, and M is usually chosen based on outcome type.

The choice of loss determines what the benefiting score estimates

For a loss M that is the negative log-likelihood under a canonical link $f(X)$ is a fixed, invertible transform of a single treatment contrast $\tau(X)$:

Outcome	Loss M	Benefiting score $f(X)$
Continuous	Squared error	$\tau_{\text{CATE}}(X)$
Binary	Logistic	$\text{artanh}(\tau_{\text{RD}}(X))$
Time-to-event	Cox partial likelihood	$\log \text{HR}(X)$

When the loss and link imply a fixed one-to-one transformation $f(x) = h\{\tau(x)\}$, constancy of $f(x)$ is equivalent to constancy of $\tau(x)$.

An importance-based statistic closes the null hypothesis gap, approximately

Fit XGBoostML and extract feature importances $I = (I_1, \dots, I_p)$. The test statistic is

$$T = \max_{1 \leq j \leq p} |I_j|.$$

Feature importance is shift-invariant, so adding a constant c to $f(X)$ has no effect on the test statistic, T .

The shift invariant statistic (approximately) extends a test built for $H_0 : \tau(x) = 0$ to $H_0 : \tau(x) = c$.

We use SHAP values as the importance measure, for clinical interpretability

SHAP (Lundberg and Lee 2017) decomposes each prediction additively into a baseline component, and individual feature contributions:

$$\hat{f}(X_i) = E_X [\hat{f}(X)] + \sum_{j=1}^p \phi_{ij},$$

where ϕ_{ij} is the SHAP value of covariate j for subject i .

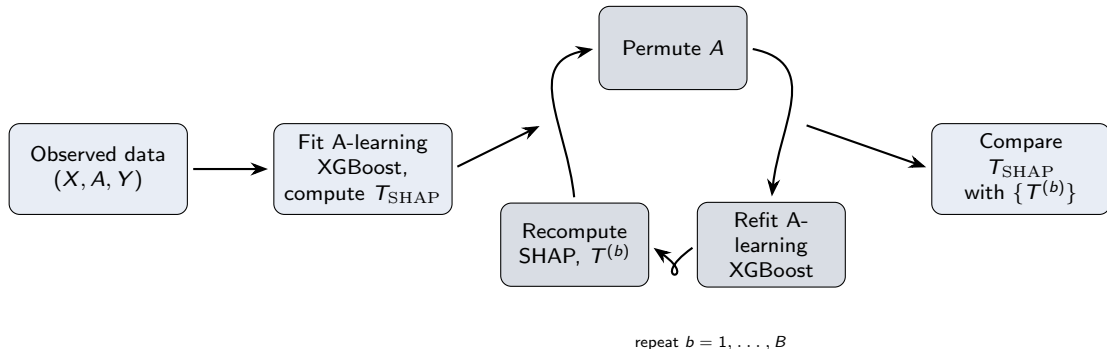
We use the mean absolute SHAP value as the covariate importance measure:

$$l_j = \frac{1}{n} \sum_{i=1}^n |\phi_{ij}| \implies T_{\text{SHAP}} = \max_{1 \leq j \leq p} l_j.$$

SHAP supplies both the global importance measure used in the test, and patient-level contributions for follow-up interpretation.

Permutation testing calibrates the full A-learning + SHAP pipeline

For T_{SHAP} , the reference distribution comes from refitting the entire pipeline under each permutation:



A-learning combined with the SHAP-based statistic lead to an approximately valid test for $H_0 : \tau(x) = c$ for all x , where c is unknown.

The A-learning test has limitations in calibration and interpretation

Calibration

- Treatment-label permutation is exact under the sharp no-effect null

$$Y_i(1) = Y_i(0) \quad \text{for every patient } i.$$

- Our scientific null is broader:

$$H_0 : \tau(x) = c, \quad \text{where } c \text{ may be non-zero.}$$

- The shift-invariant SHAP statistic targets covariate-driven variation, but does not make the permutation distribution exact for $c \neq 0$.

Interpretation

- The form of $f(X)$ depends on the outcome-specific loss, and may not be the clinical contrast of interest $\tau(X)$.

Pseudo-outcomes let us choose the treatment-effect scale first

Choose the treatment-effect contrast $\tau(x)$ and construct a pseudo-outcome D satisfying

$$E(D \mid X = x) = \tau(x).$$

We then fit regular XGBoost directly to D :

$$D \longrightarrow \hat{\tau}(X).$$

SHAP explains the fitted treatment-effect contrast on the $\tau(x)$ scale.

The pseudo-outcome turns estimation of $\tau(x)$ into a regression problem.

For a mean-difference target, a natural pseudo-outcome is doubly-robust

Let

$$m_a(X) = E(Y \mid A = a, X).$$

An augmented inverse-probability-weighted pseudo-outcome is

$$D_i = \hat{m}_1(X_i) - \hat{m}_0(X_i) + \frac{A_i}{\hat{\pi}(X_i)} \{Y_i - \hat{m}_1(X_i)\} - \frac{1 - A_i}{1 - \hat{\pi}(X_i)} \{Y_i - \hat{m}_0(X_i)\}.$$

The construction is doubly robust: it is valid if either the outcome models $m_a(X)$ or the propensity score model is correct.

In an RCT, $\pi(X)$ is known from the randomization design and is therefore correctly specified.

For a continuous outcome, a candidate constant effect gives a null construction

Suppose the target is the conditional mean difference and consider

$$H_0 : \tau(x) = c \quad \text{for every } x.$$

To construct a permutation distribution for a candidate value c , impose the stronger sharp additive null

$$Y_i(1) - Y_i(0) = c \quad \text{for every patient } i.$$

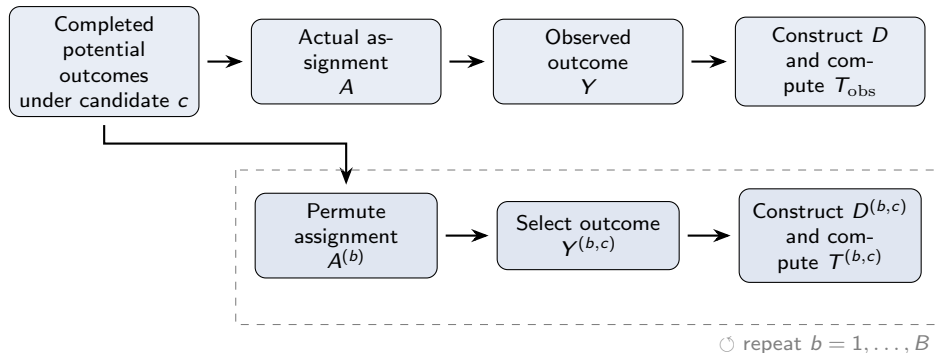
The missing potential outcome can then be imputed as

$$A_i = 1 \quad : \quad Y_i(0) = Y_i - c,$$

$$A_i = 0 \quad : \quad Y_i(1) = Y_i + c.$$

The conditional-mean null does not imply the sharp additive null. This is an additional approximation used to construct the permutation distribution.

The pseudo-outcome permutation procedure generates null data first



The fixed- c permutation distribution is

$$\{T^{(1,c)}, \dots, T^{(B,c)}\}.$$

The Berger and Boos (1994) approach accounts for the unknown constant, c

The null contains an unknown value c :

$$H_0 : \tau(x) = c.$$

For each fixed c , obtain a permutation p -value $p(c)$. Let $\mathcal{C}$ be a $(1 - \gamma)$ confidence set for the common effect.

The Berger–Boos adjusted p -value is

$$p_{\text{BB}} = \sup_{c \in \mathcal{C}} p(c) + \gamma.$$

Rather than plug in one estimate of c , the procedure considers all plausible values in $\mathcal{C}$ (in practice, a fixed grid).

The pseudo-outcome and null-imputation steps depend on the outcome type

Outcome	DR pseudo-outcome	Null imputation
Continuous	Mean difference	Sharp additive null
Time-to-event	RMST difference (with IPCW)	No direct event-time imputation
Binary	Risk difference	No additive shift for $c \neq 0$

Pseudo-outcomes extend across outcome types; the null-imputation construction does not.

Comparing the approaches: simulation data-generating process

Generate covariates

$$X \sim \mathcal{MVN}_{10}(0, \Sigma), \quad \Sigma \text{ equicorrelated.}$$

Continuous:

$$Y(A) = s [-1.5 + \{\mathbf{1}(X_1 < 0) + \mathbf{1}(X_2 < 0)\}A] + \beta_0(X_3 + X_4) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1).$$

Binary:

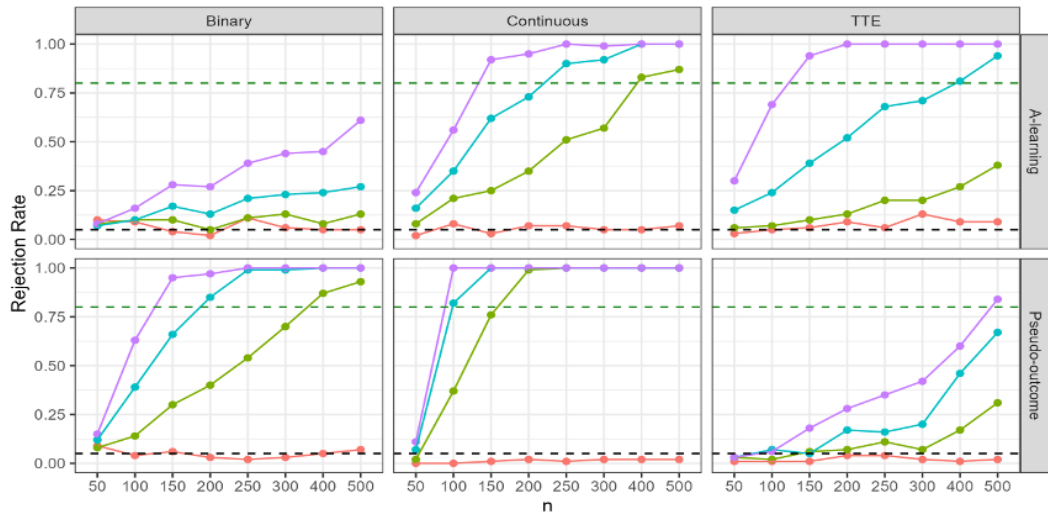
$$Y_b(A) = \mathbf{1}\{Y(A) \geq 0\}$$

Time-to-event:

$$Y_t(A) = \min[A \exp\{Y(1)\} + (1 - A) \exp\{Y(0)\}, C], \quad C \sim \text{Lognormal}(0, 3).$$

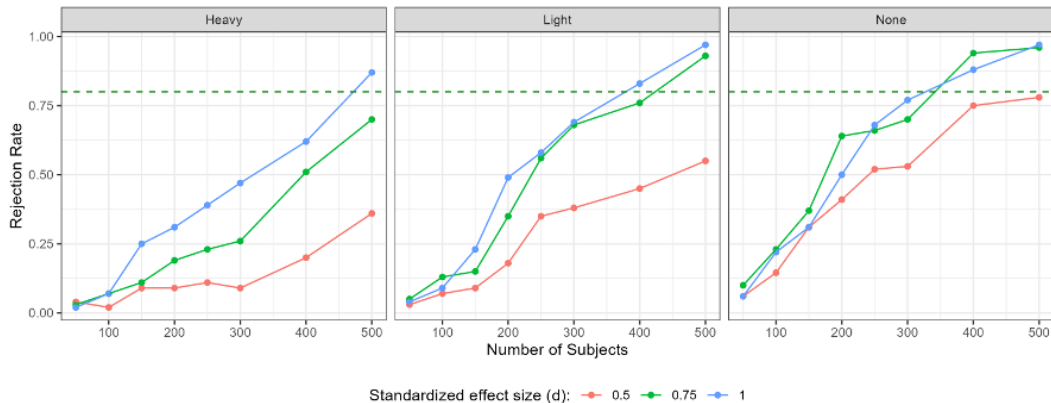
We vary s across outcome types to target a standardized effect size d .

Power curves under the A-learning and pseudo-outcome approaches



Standardized effect size (d): 0 0.5 0.75 1

Performance for the time-to-event pseudo-outcome approach depends on censoring



Summary

- **A-learning**: one framework, with outcome-specific loss functions; validity for $H_0 : \tau(x) = c$ is only approximate, and the benefiting score may not equal the clinical contrast of interest.
- **Pseudo-outcomes**: targets $\tau(x)$ directly and gives a better test in the continuous case, at the cost of an outcome-specific construction and typically lower power for time-to-event outcomes.
- **SHAP**: in both cases, SHAP provides more interpretable patient-level summaries, and follow-up analyses.

Neither approach dominates: A-learning offers generality and a unified pipeline; pseudo-outcomes offer direct clinical interpretability.

References



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For time-to-event outcomes, RMST is a clinically interpretable target

The previous construction targeted a conditional mean difference. For a time-to-event outcome, we use the RMST difference.

Let $T(a)$ denote the event time under treatment a and let

$$S_a(t \mid X = x) = P\{T(a) > t \mid X = x\}$$

denote the corresponding survival function. For a horizon L ,

$$\text{RMST}_a(x) = E\{\min(T(a), L) \mid X = x\} = \int_0^L S_a(t \mid X = x) dt.$$

The treatment-effect target is

$$\tau_{\text{RMST}}(x) = \text{RMST}_1(x) - \text{RMST}_0(x).$$

RMST measures expected time alive and event-free through time L .

How SHAP values are computed

For covariate j , the Shapley value averages its marginal contribution across all coalitions S of the other covariates:

$$\phi_{x_{ij}} = \sum_{S \subseteq P \setminus \{j\}} \frac{|S|! (p - |S| - 1)!}{p!} \left(f(X_{i, S \cup \{j\}}) - f(X_{i, S}) \right).$$

This is the unique attribution satisfying efficiency, symmetry, dummy, and additivity — efficiency is what gives the decomposition $f(X_i) = E_X[f(X)] + \sum_j \phi_{x_{ij}}$.

Direct computation is exponential in p . **TreeSHAP** (Lundberg, Erion, et al. 2020) computes these values exactly, in polynomial time, by exploiting the tree structure directly.